MIL-S-16788A
29 March 1971
SUPERSEDING
MIL-S-16788
13 November 1951

MILITARY SPECIFICATION

STEEL, CARBON, HOT-ROLLED OR FORGED (BLOOMS, BILLETS, AND SLABS FOR REFORGING)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hot-rolled or forged carbon steel semifinished products (blooms, billets, and slabs) for reforging. (See 6.1.)

1.2 Classification. The carbon steel covered by this specification shall be furnished in the following classes, designated according to chemical compositions (see table I), as specified (see 6.2):

Class C1
Class C2
Class C3
Class C4
Class C5
Class C6
Class C8
Class C10

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 9640

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-430	Macrograph Standards for Steel Bars, Billets, and Blooms

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability
FED-STD-151	Metals; Test Methods

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM-317	Macroetch Testing and Inspection of Steel Forgings

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Process. The steel shall be made by either the electric furnace, open hearth, or basic oxygen process. Steel shall be cast in metal ingot molds. The use of hot tops of any suitable material is permitted.

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3.2 Discard. Sufficient discard shall be made from each ingot to insure freedom from injurious piping and undue segregation.

3.3 Reduction from ingot. Unless otherwise specified, the material shall be made from ingots having at least two times the cross-sectional area of the material.

3.4 Chemical composition.

3.4.1 Ladle analysis. The ladle analysis of the steel shall conform to the chemical composition requirements shown in table I. The contractor shall furnish a report of the chemical analysis of each heat of steel.

Table I

CHEMICAL COMPOSITION (PERCENT)[1]

Class	Carbon	Manganese	Phosphorus (maximum)	Sulphur (maximum)
C1	0.08 - 0.13	0.30 - 0.60	0.040	0.050
C2	0.18 - 0.23	0.30 - 0.60	0.040	0.050
C3	0.28 - 0.34	0.60 - 0.90	0.040	0.050
C4	0.37 - 0.44	0.60 - 0.90	0.040	0.050
C5	0.48 - 0.55	0.60 - 0.90	0.040	0.050
C6	0.55 - 0.65	0.60 - 0.90	0.040	0.050
C8	0.75 - 0.88	0.60 - 0.90	0.040	0.050
C10	0.90 - 1.03	0.30 - 0.50	0.040	0.050

[1] For commercial chemical designations, see 6.3.

3.4.2 Check analysis. The chemical composition of blooms, billets, or slabs shall conform to the requirements specified in table I, subject to check analysis tolerances of FED-STD-66.

3.5 Defects. All material shall be free from cracks or other imperfections that may affect its usability.

3.5.1 Macroetch. When specified by the procuring activity, the material shall be subject to a macroetch test in accordance with 4.3.3. The quality of the steel, as indicated by the results of the macroetch

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test for sections up to and including 36 square inches, shall be equal to, or better than, macrograph conditions S4, R4, and C4. For sections over 36 square inches, macrograph conditions S5, R5, and C5 are considered acceptable limits as classified in MIL-STD-430.

3.6 Conditioning. Unless otherwise specified, the maximum permissable depth of conditioning for removal of injurious surface defects, provided the width of the conditioning is at least four times the greatest conditioning depth, shall be as follows.

3.6.1 The depth of conditioning shall not exceed one-sixteenth inch for each inch of material thickness when the material width is less than twice the material thickness.

3.6.2 When the material width is equal to or greater than twice the material thickness, the depth of conditioning shall not exceed three thirty-seconds of an inch for each inch of material thickness.

3.6.3 In either case (3.6.1 or 3.6.2), the maximum permissible depth of conditioning shall not exceed three-fourths inch.

3.6.4 When conditioning is required on both opposite locations of parallel sides, the maximum permissible combined depth of conditioning shall not exceed 1-1/8 inches.

3.7 Dimensions . The nominal dimensions of the material shall be as specified by the procuring activity.

3.8 Weight. Unless otherwise specified by the procuring activity, the permissible variations from the specified or theoretical weight of less than carload lots shall be within ±5 percent. For carload lots, the variation shall be within ±2-1/2 percent of the total lot weight.

3.9 Heat treatment. Unless otherwise specified by the procuring activity, classes C6, C8, and C10 shall be supplied in the annealed condition.

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3.10 Latent imperfections. Material containing injurious imperfections which are not manifest upon the original inspection, but develop during forging or machining operations, may upon verification be returned to the contractor for replacement or as otherwise mutually agreed upon.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lots.

4.2.1.1 Melting furnace heat identified. If material is positively identified as to melting furnace heat, a lot shall consist of all material made therefrom.

4.2.1.2 Melting furnace heat not identified. If material is not identified as to melting heat, a lot shall consist of all material of the same cross section size.

4.2.2 Samples.

4.2.2.1 Chemical analysis. Sampling for chemical analysis shall be in accordance with table II, with tolerances in accordance with FED-STD-66.

4.2.2.2 Macroetch. When macroetch testing is required, the sampling procedure shall be as specified in 4.2.2.2.1 or 4.2.2.2.2.

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Table II

SAMPLING FOR CHEMICAL ANALYSIS

Number of pieces in lot	Number of samples required	
	Lot as defined in 4.2.1.1	Lot as defined in 4.2.1.2
1	1	1
2	2	2
3 - 30	3	3
More than 30	3	10 percent of number of pieces in lot

4.2.2.2.1 Identified material. If material can be identified as to ingot and heat, samples, representing the entire cross-sectional area and approximately one-half inch in thickness, shall be cut from the top of the top billet and the bottom of the bottom billet of the first, middle, and last usable ingot of heat.

4.2.2.2.2 Unidentified material. If material cannot be identified as to ingot or heat, samples, representing the entire cross-sectional area and approximately one-half inch in thickness, shall be cut from each end of each piece of material. On sections larger than 10 inches in diameter or 14 inches diagonal, a segment representing a cross-sectional surface from center to the outside may be used. Where inspection is to be made of pieces which are to be cut into shorter lengths prior to delivery, slices from the ends prior to cutting are sufficient.

4.3 Tests.

4.3.1 General procedures. General testing procedures shall be accomplished in accordance with FED-STD-151.

4.3.2 Chemical analysis. Samples shall be subject to a chemical analysis (see 3.4.2) at a laboratory designated by the bureau or agency concerned.

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4.3.3 Macroetch. When macroetching is specified by the procuring activity, testing shall be accomplished in accordance with ASTM 317.

4.3.4 Defective material. Where requirements as to length permit, a defective end may be cut off and a new test or tests made until sound metal is reached. Sufficient metal shall be removed to assure removal of the defects disclosed by the tests.

4.3.5 Criteria of acceptability. Any sample subjected to the examinations and tests of section 4 which fails to meet the requirements of section 3 shall result in rejection of the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packing. Material shall be prepared for shipment in accordance with MIL-STD-163.

5.2 Marking. Shipping containers or packages shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The semifinished material covered by this specification is intended for reforging uses. This specification does not include requirements for hot-rolled bar stock.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Class required (see 1.2)
(c) Dimensions (see 3.7)
(d) Macroetch tests, when required (see 3.5.1)
(e) Permissable variation in weight, if other than 3.8.

6.3 Commercial chemical designations. Table III lists, for information only, commercial chemical designations equivalent to the compositions given in 1.2.

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Table III

COMMERCIAL CHEMICAL COMPOSITION

Class (this specification)	Commercial equivalent	
	AISI	SAE
C1	C1010	1010
C2	C1020	1020
C3	C1030	1030
C4	C1040	1040
C5	C1050	1050
C6	C1060	1060
C8	C1080	1080
C10	C1095	1095

Custodians:
Army - MR
Navy - OS
Air Force - 84

Review interest:
Navy - SH

User activities:
Army - MU

Preparing activity:
NAVY - OS
(Project No. 9640-0009)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-16788A, STEEL, CARBON, HOT-ROLLED OR FORGED (BLOOMS, BILLETS, AND SLABS FOR REFORGING)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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